Basic Principles of Medicine 1
Module: Foundations to Medicine
Lecture No: 13
Title: Flipped class: X-linked Inheritance

Dr. Shellon Thomas

sthoma10@sgu.edu

Acknowledgement: Dr. Sharmila Upadhya

Required pre-reading: DLA: X-linked inheritance

- DLA 12a: X-linkage examples
- DLA 12b: Unexpected findings in Mendelian inheritance (Shown with examples)

SOM.MK.III.BPM1.1.FTM.3.GNET.0037	Distinguish the mode of inheritance, identify the main clinical features, and interpret genetic principles illustrated by the following X-linked disorders: Hemophilia A and B, G6PD deficiency, Duchenne and Becker muscular dystrophy, red/green color blindness. Lesch Nyhan syndrome, X-SCID as examples of X-linked disorders
SOM.MK.III.BPM1.1.FTM.3.GNET.0038	Summarize and distinguish the characteristics of the following patterns of inheritance using pedigrees: autosomal dominant, autosomal recessive, X-linked dominant and X-linked recessive
SOM.MK.III.BPM1.1.FTM.3.GNET.0039	Demonstrate how disease liability is transmitted for X-linked diseases
SOM.MK.III.BPM1.1.FTM.3.GNET.0040	Propose how the basic risk calculations are performed for X-linked disorders
SOM.MK.III.BPM1.1.FTM.3.GNET.0041	Explain how a woman who is a carrier of an X-linked disorder might manifest symptoms of the condition (manifesting heterozygote) Skewed X-inactivation, non-random X–inactivation
SOM.MK.III.BPM1.1.FTM.3.GNET.0042	Discuss X-linked dominant disorders and traits
SOM.MK.III.BPM1.1.FTM.3.GNET.0043	Compare and contrast variable expressivity and incomplete penetrance; Give examples of each
SOM.MK.III.BPM1.1.FTM.3.GNET.0044	Describe pleiotropy
SOM.MK.III.BPM1.1.FTM.3.GNET.0045	Compare and contrast locus heterogeneity and allelic heterogeneity; gve examples
SOM.MK.III.BPM1.1.FTM.3.GNET.0046	Describe locus heterogeneity in terms of complementation groups
SOM.MK.III.BPM1.1.FTM.3.GNET.0047	Explain how allelic heterogeneity allows the formation of a compound heterozygote in autosomal recessive disorders

Objectives: Flipped class: X-linked inheritance

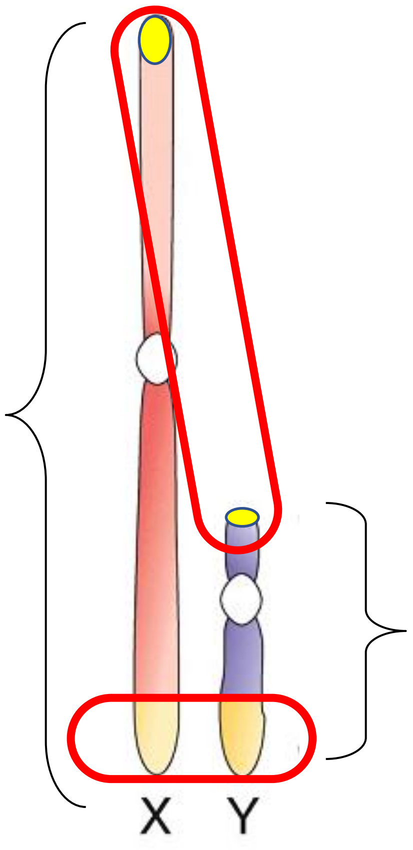
SOM.MK.III.BPM1.1.FTM.3.GNET.0048	Demonstrate how disease liability is transmitted for autosomal dominant diseases, autosomal recessive diseases and X-linked diseases
SOM.MK.III.BPM1.1.FTM.3.GNET.0049	Distinguish the different characteristics of recessive and dominant X-linked traits using pedigrees
SOM.MK.III.BPM1.1.FTM.3.GNET.0050	Identify obligate carriers in a pedigree depicting an X-linked trait
SOM.MK.III.BPM1.1.FTM.3.GNET.0051	Compare and contrast between X-linked, Y-linked, AR, AD, and mitochondrial traits.
SOM.MK.III.BPM1.1.FTM.3.GNET.0052	Discuss the concept of penetrance and variable expression and Differentiate between locus and allelic heterogeneity
SOM.MK.III.BPM1.1.FTM.3.GNET.0053	Calculate the recurrence risk based on penetrance for autosomal dominant and autosomal recessive disorders
SOM.MK.III.BPM1.1.FTM.3.GNET.0054	Define pleiotropy with examples. Be able to identify concepts if given a clinical scenario
SOM.MK.III.BPM1.1.FTM.3.GNET.0055	Recognize occurrence of new mutations as a cause of new genetic diseases in families. Be able to identify this concept from clinical scenarios and pedigrees
SOM.MK.III.BPM1.1.FTM.3.GNET.0056	Explain the phenomenon of germinal (germ-line) mosaicism
SOM.MK.III.BPM1.1.FTM.3.GNET.0057	Identify genetic disorders that have a delayed age of onset, and be able to identify this concept from clinical scenarios and pedigrees
SOM.MK.III.BPM1.1.FTM.3.GNET.0058	Explain the phenomenon of a manifesting heterozygote in X-linked recessive disorders, and be able to identify this concept from clinical scenarios and pedigrees

Recommended Reading

- Human Genetics and Genomics, 4th Edition Korf and Irons Chapter 3.
  - You can find the above book page in your Sakai site under:
  - Resources/General Information/Book Access

Sex Chromosomes

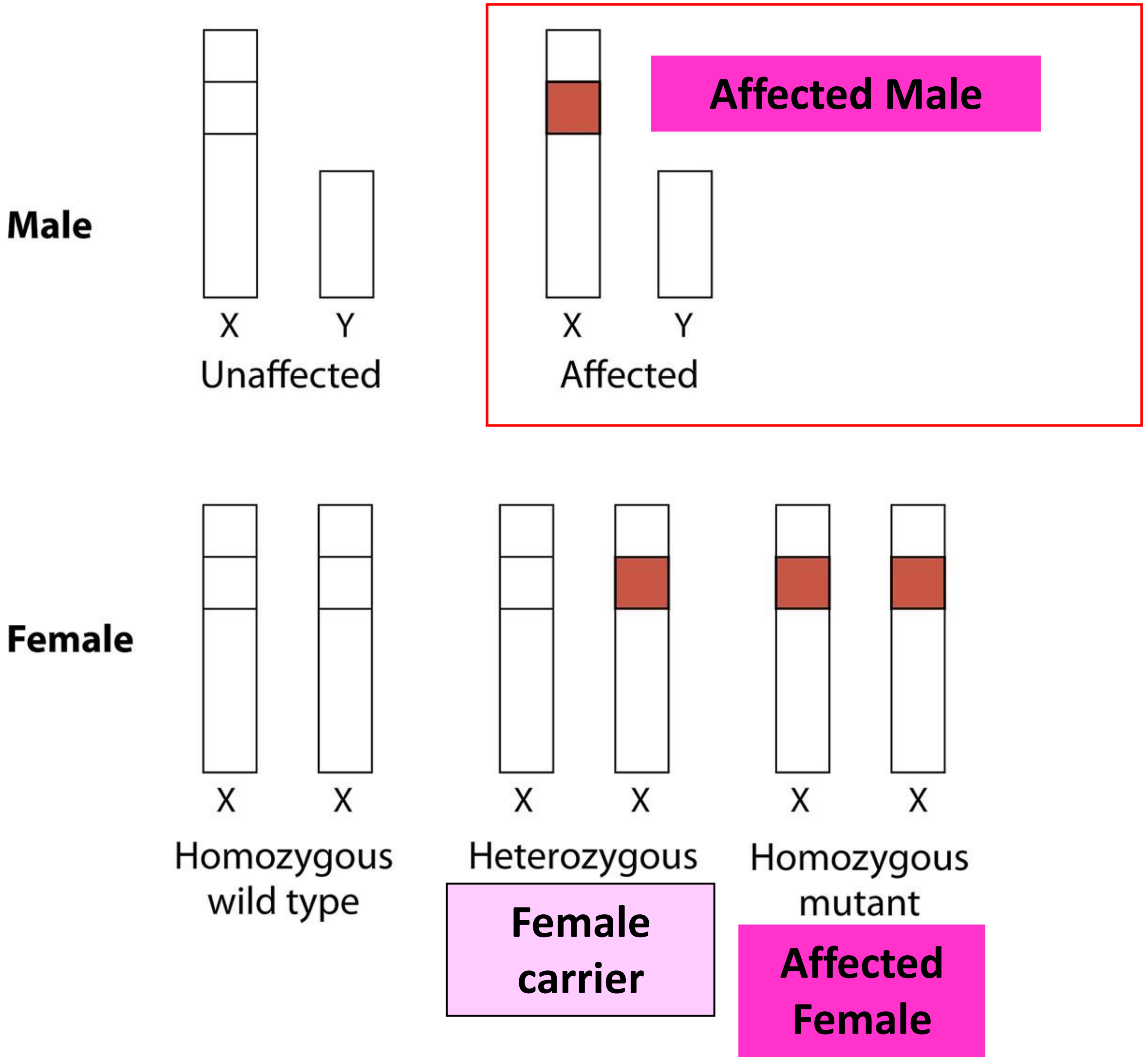
- 1095 genes are annotated on the X chromosome (this is the current estimation; maybe there are more)
- **Females have two copies of all genes on the X-chromosome**
- **Males are said to be <u>hemizygous</u> for the X chromosome (males only have one copy)**



About 148 genes mapped to the Y chromosome (mostly related to spermatogenesis)

Pseudoautosomal region: regions of X and Y chromosomes that match. It is necessary to line up chromosomes correctly during meiotic recombination

X-Linked Recessive Disorders



Females homozygous for X-linked disorders are typically very rare in population

Human Genetics and Genomics, Fourth Edition. Bruce R. Korf and Mira B. Irons.

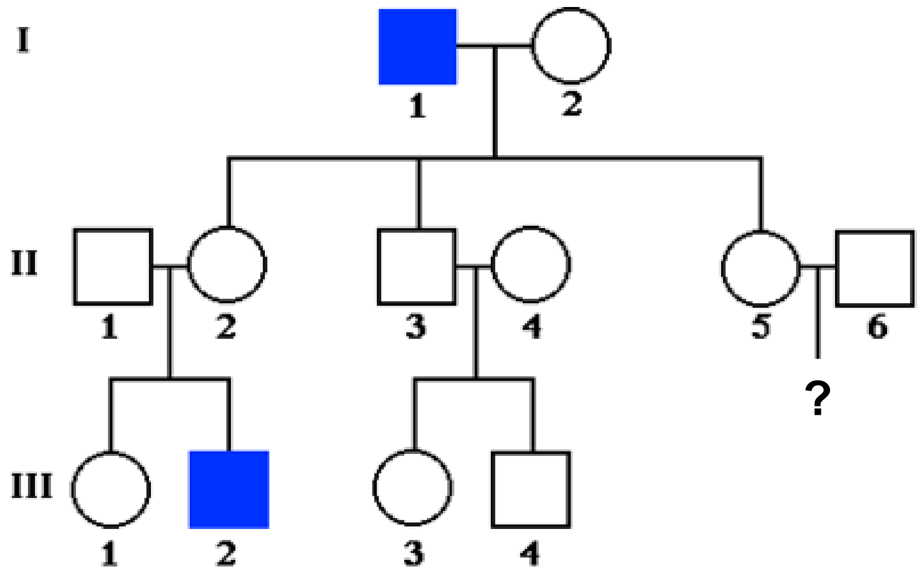
Figure 3.14 Males who carry a mutation on the X chromosome will express the phenotype associated with that mutation. Females can be homozygous for either the wild type or the mutant allele or can be heterozygous. Whether they express the mutant allele depends on whether it behaves as a recessive or a dominant trait.

Clicker Question Next!

Please prepare to answer
the clicker question
on the next slide.

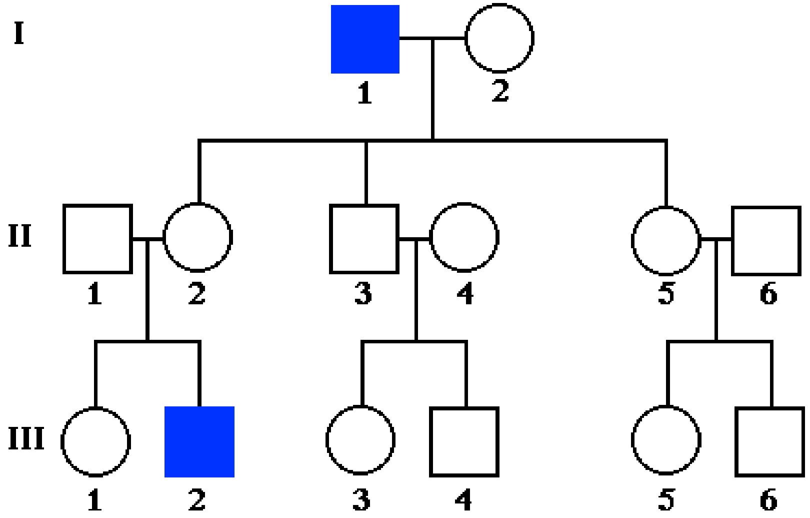
Individual II-5 comes to the physician after learning that she is pregnant. Some individuals in her family are affected with a bleeding disorder as shown in the pedigree. What best describes her risk of having an affected child?

A.	100%	14%
B.	67%	14%
C.	50%	14%
D.	33%	14%
E.	25%	14%
F.	Population risk	14%
G.	0%	14%



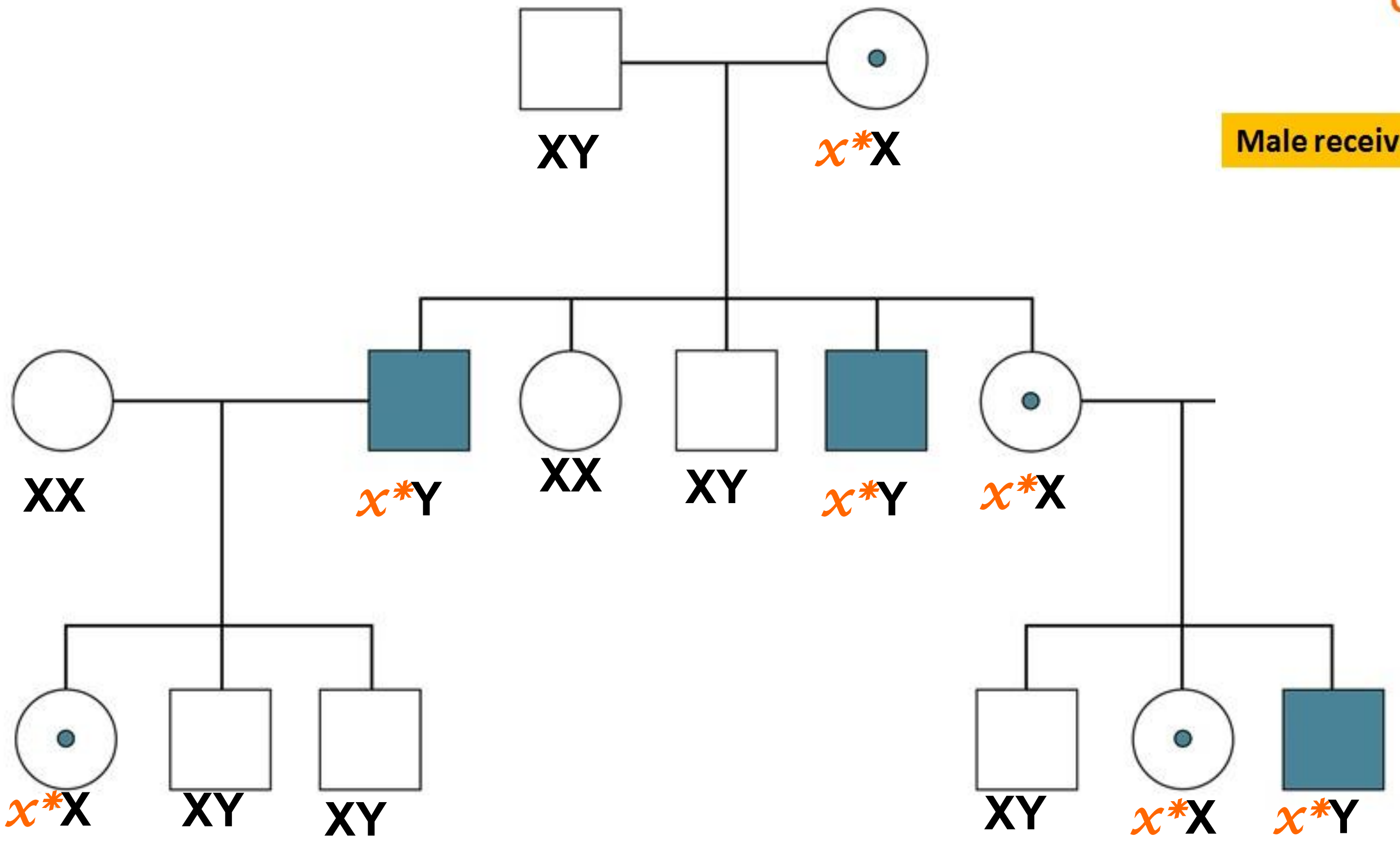
72

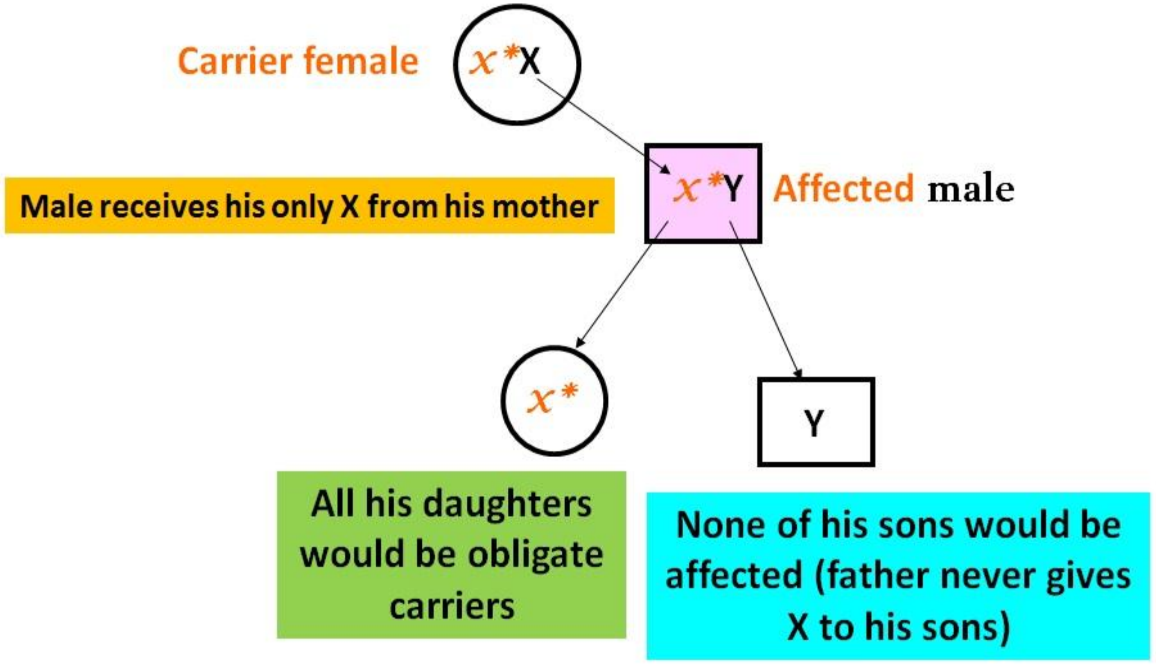
X – Linked Recessive Disorders



- Males require only one copy of the mutation (hemizygous) to express the disease.
- Disorders are more common and more severe in **males** than in females
- Skipped generations common
- Affected father transmits the mutation to all his daughters who are typically NOT affected but they are **carriers**. A carrier daughter may transmit the mutation to her sons who would then be affected.
- Male to male transmission is not seen

Transmission of an X-linked recessive disorder





Human Genetics and Genomics, Fourth Edition. Bruce R. Korf and Mira B. Irons.

Figure 3.15 Pedigree illustrating X-linked recessive transmission. Note the absence of male-to-male transmission.

Recurrence risk for X-linked recessive disorder
Normal Father (XY) x Carrier mother (Xx*)

	Carrier mother	
Normal father	x*	X
X	x*X	XX
Y	x*Y	XY

Pathogenic allele

On average…

50% of daughters may be carriers; 50% of daughters may be normal

50% of sons may be affected; 50% of sons may be normal

Each conception event is independent

SOM.MKIII.BPM1.1.FTM.3.GNET.0048

SOM.MKIII.BPM1.1.FTM.3.GNET.0048 SOM.1a.BPM1.1.FTM.3.GNET.GN 0403

Recurrence risk for X-linked recessive disorder
Affected Father (x^*Y) x Normal mother (XX)

	Normal mother	
Affected father	X	X
x^*	x^*X	x^*X
Y	XY	XY

Mutant allele → x^*

All daughters of an affected father are <u>obligate</u> carriers of the mutant allele

A son may not inherit an X-linked mutation from his father

SOM.MKIII.BPM1.1.FTM.3.GNET.0048

Recurrence risk for X-linked recessive disorder
Affected Father (x*Y) x Carrier mother (x*X)

This is a relatively rare occurrence in population – usually seen for non-lethal X-linked traits (like red-green color blindness) where the carrier frequency is relatively high in the population, and males with the trait are not severely affected.

	Carrier mother	
Affected father	x*	X
x*	x* x*	x▫X
Y	x*Y	XY

Mutant allele

Each conception is independent event, **but on average**:

50% of daughters may be carriers;
50% of daughters may be affected

50% of sons may be affected;
50% of sons may be normal

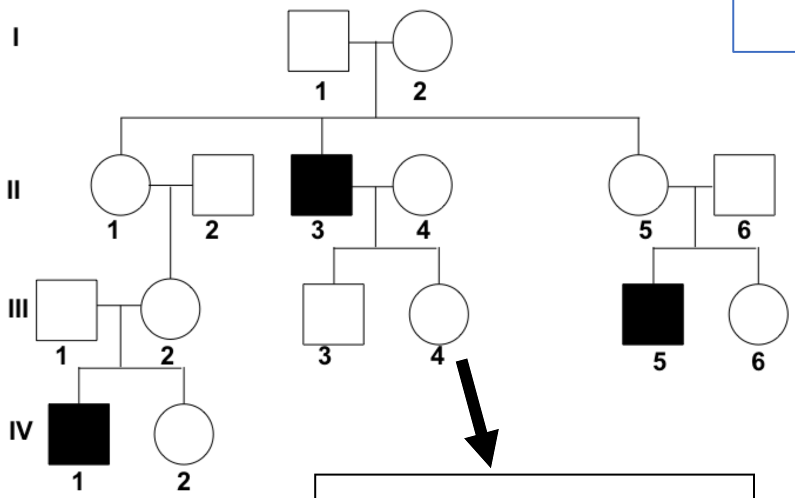
Clicker Question Next!

Please prepare to answer
the clicker question
on the next slide.

Individual III-4 comes to the physician after learning that she is pregnant. Some of her family members have inherited clotting factor VIII deficiency as shown in the pedigree. Ultrasound shows that she is carrying a male fetus. What is the best description of the risk that this child will be affected?

A. 100%	14%
B. 67%	14%
C. 50%	14%
D. 33%	14%
E. 25%	14%
F. Population risk	14%
G. 0%	14%

Identify obligate carriers



Ultrasound shows a male fetus

:72

SOM.1a.BPM1.1.FTM.3.GNET.0048

X-Linked Recessive Disorders

- Dystrophin associated muscular dystrophy
  - Duchenne muscular dystrophy (severe)
  - Becker muscular dystrophy (milder form)
  - both are due to mutations of the same gene (*DMD*, dystrophin)

- Hemophilia A and B result in bleeding tendencies

- Glucose 6-phosphate dehydrogenase (*G6PD*) deficiency
  - hemolytic anemia on ingestion of primaquine, sulfa drugs
- Lesch-Nyhan syndrome [Hypoxanthine Guanine Phospho ribosyl transferase (*HGPRT*) deficiency]
  - causes hyperuricemia, gout, & self mutilation

X-linked enzyme deficiency disorders

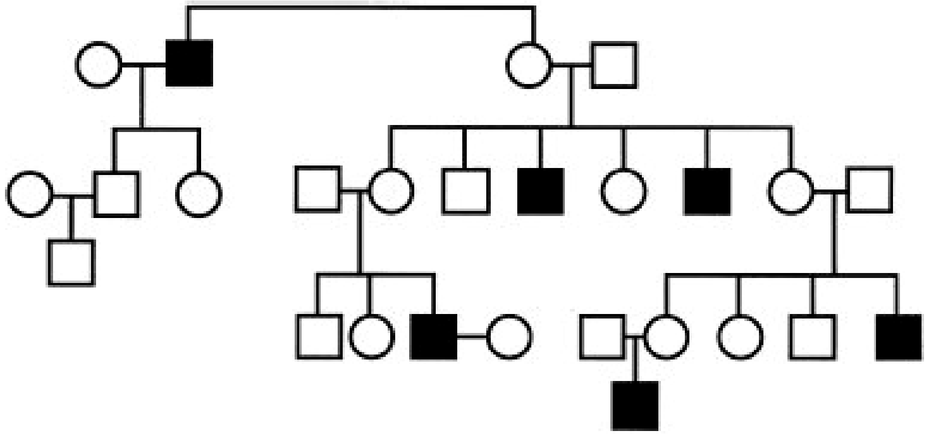
- Red-green color blindness/ deficiency (non-lethal)

- X-linked SCID (*IL2RG* gene)

Clicker Question Next!

Please prepare to answer
the clicker question
on the next slide.

Which of the following disorders is most likely segregating in the family depicted by the pedigree?



72

A. Duchenne muscular dystrophy 20%
B. Phenylketonuria 20%
C. Becker muscular dystrophy 20%
D. Huntington disease 20%
E. Myotonic dystrophy 20%

The dystrophin gene is involved in both DMD and BMD (same gene; variable expressivity). Boys with Duchenne muscular dystrophy are typically too sick to have children– see **GNET DLA**

A scientist studying a family that has several individuals affected with a genetic disorder constructs the pedigree shown in the figure. Which of the following is the best probability estimate that the indicated individual is a carrier of the mutation?

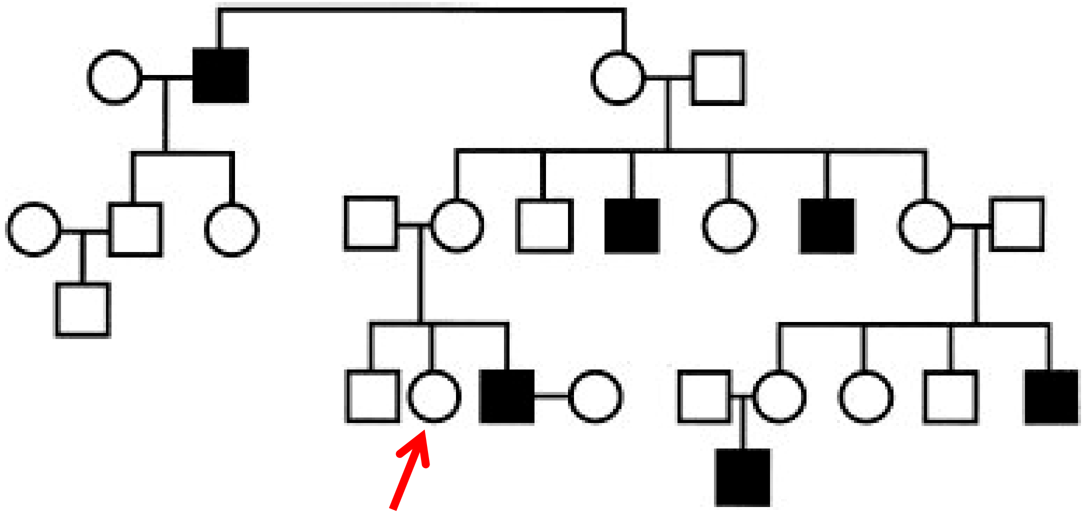
A. ½ 20%

B. ¼ 20%

C. 1/8 20%

D. 1 20%

E. 2/3 20%



	X	x*
X	XX	Xx*
Y	XY	x*Y

Strategy:

First - Identify the mode of transmission.

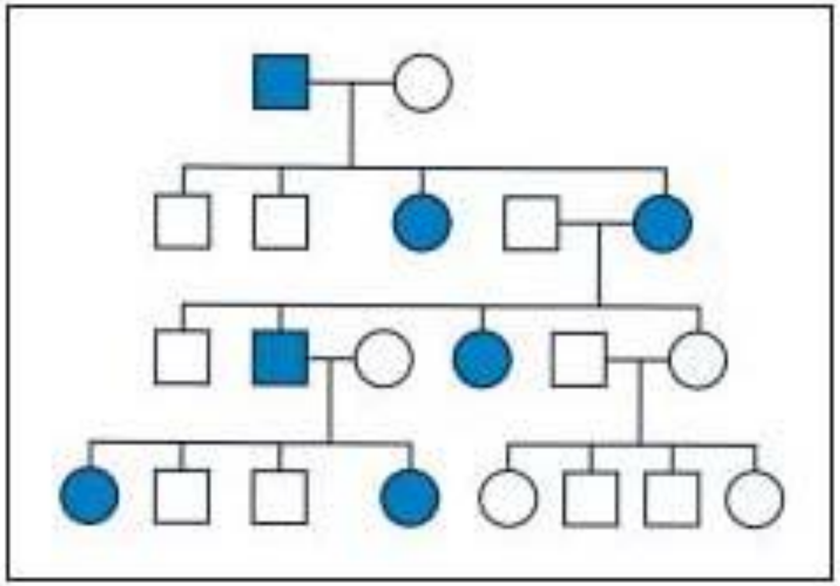
Then: Identify all the obligate carriers of the mutant allele (II-5 is carrier). Daughters of carrier mothers have a 50% carrier risk

72

From GNET DLA

SOM.MKIII.BPM1.1.FTM.3.GNET.0050

X-Linked Dominant Disorders (or traits)



Females would be affected more often for two reasons:

- Females are diploid for the X chromosome; so they are twice as likely to have an X chromosome with the mutation compared to a male.
- Many of the X-linked dominant disorders are very (very) severe in males (because males are hemizygous).

- Skipping of generations not common
- Preponderance of females compared to males
- No male to male transmission
- Affected male transmits the disease to **all his daughters**, but none of his sons would be affected
- Variable expressivity in females: Due to the phenomena of X-inactivation
- Males are often very severely affected (sometimes lethal)

Recurrence risk for X-linked dominant disorder (or traits)
Affected Father (X*Y) x Normal mother (XX)

	Normal mother	
Affected father	X	X
X*	X* X	X* X
Y	XY	XY

Chromosome with mutant allele → X*

All daughters have the mutant allele and are affected

All sons are normal

Recurrence risk for X-linked dominant disorder
Normal Father (XY) x Affected mother (XX*)

	Affected mother	
Normal father	X*	X
X	X* X	XX
Y	X*Y	XY

Chromosome with mutant allele

On average…

- **50% chance daughters will inherit the allele and be affected**
- **50% chance to be normal**

50% of sons may be affected and 50% of sons may be normal

Clicker Question Next!

Please prepare to answer
the clicker question
on the next slide.

A 32-year-old woman is brought to the emergency department because of stupor and confusion, coincident with a SARS CoV-2 infection. Laboratory tests show elevated blood ammonia, low blood urea, and orotic aciduria. She denies drug and alcohol use, and other toxin tests are negative. Her brother and her maternal uncle both have a urea cycle defect due to X-linked *OTC* deficiency (ornithine transcarbamoylase). Which of the following best explains the pathology in this woman?

A. Genetic anticipation — 20%
B. Classic autosomal dominance — 20%
C. Pseudoautosomal dominance — 20%
D. Nonrandom X-inactivation — 20%
E. Allelic heterogeneity — 20%

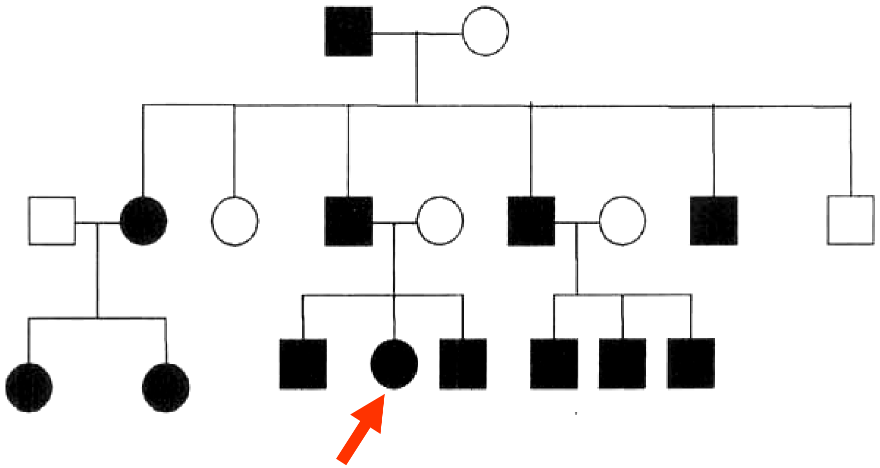
72

Manifesting heterozygote due to skewed X-inactivation
Notice that you don't need to know what OTC is, as the vignette gives that it is X-linked

From GNET DLA

A scientist identifies a family that has many individuals affected by a genetic disorder that follows typical Mendelian inheritance. Identify the most likely inheritance pattern of this disorder. Also, what is the risk that the offspring of the indicated individual would be affected if she marries a person with the normal alleles of the gene?

A. Autosomal dominant, 100%	17%
B. Autosomal recessive, 25%	17%
C. X linked recessive, 25%	17%
D. Autosomal dominant, 50%	17%
E. X linked recessive, 50%	17%
F. X-linked dominant, 50%	17%

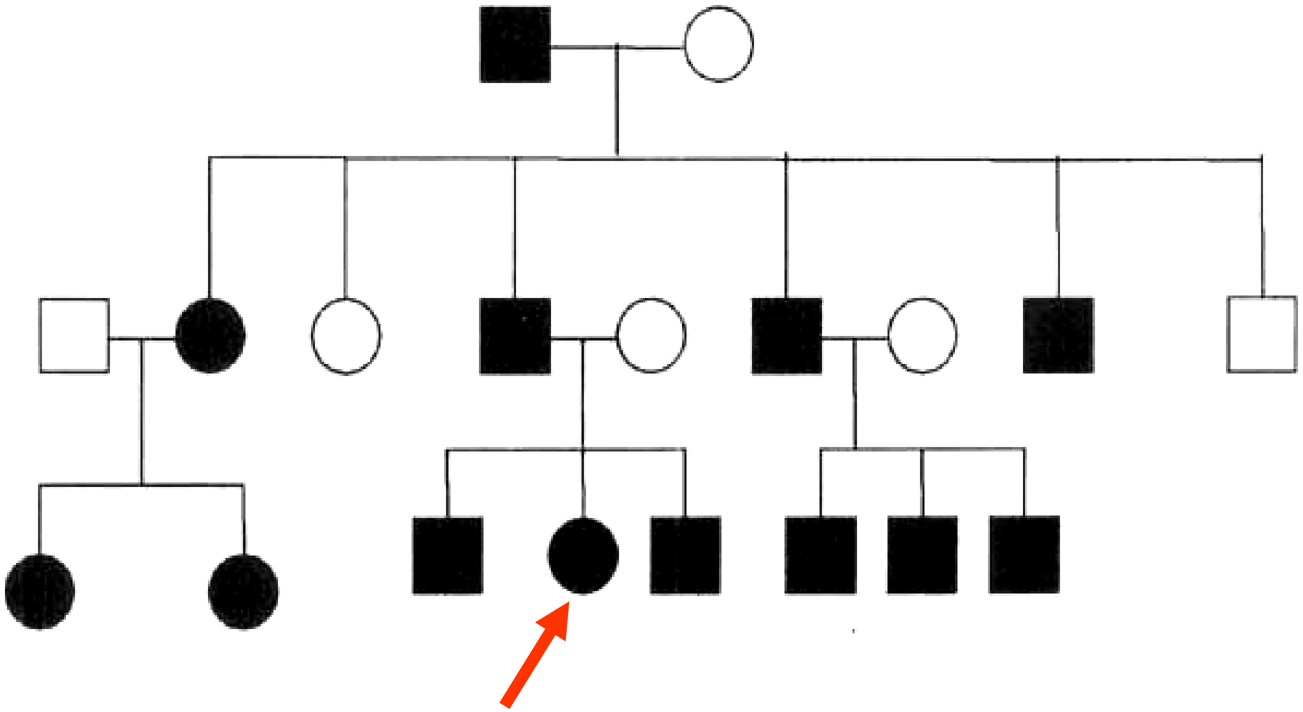


From GNET DLA

72

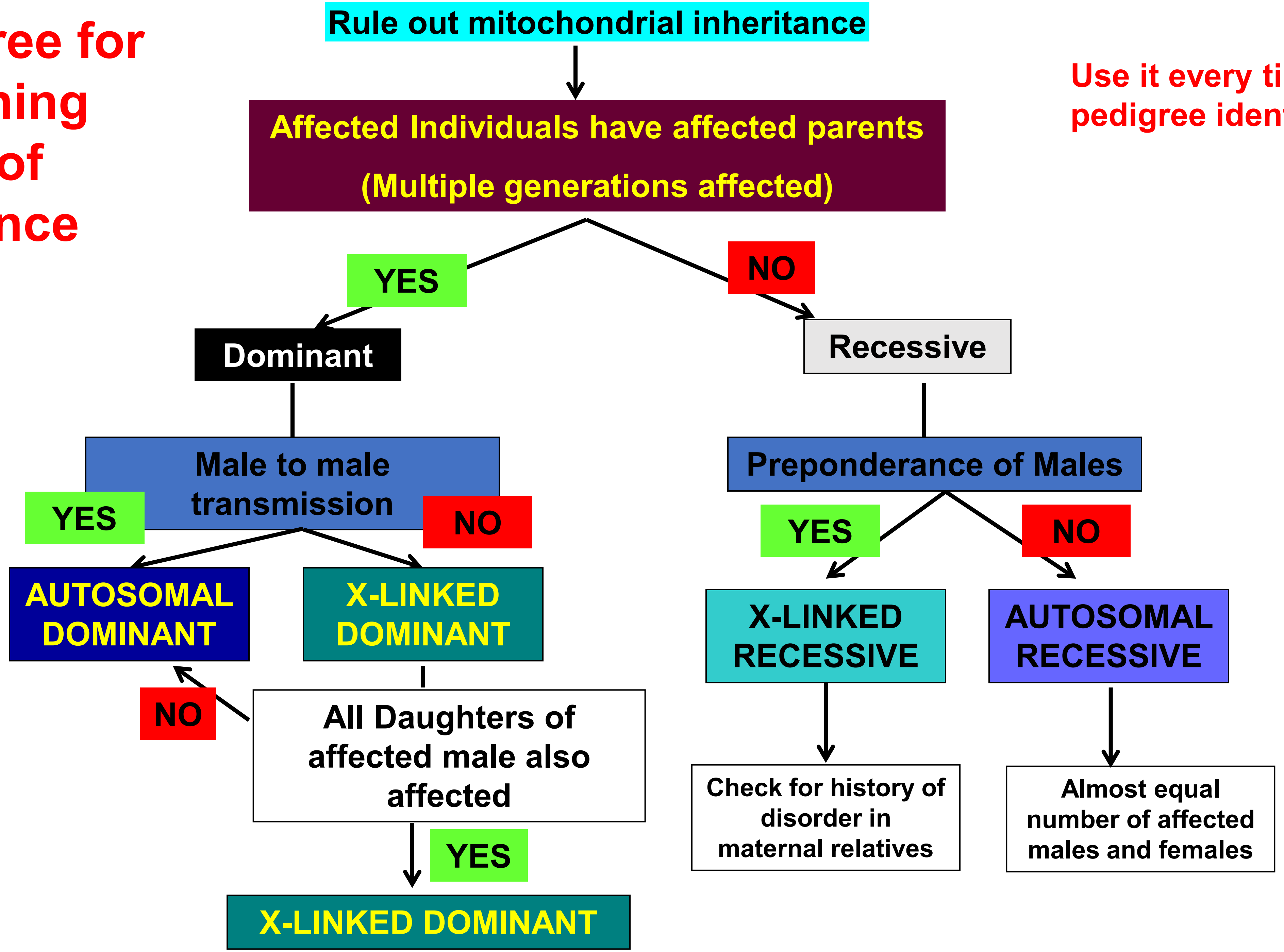
Autosomal dominant inheritance

- Note the vertical inheritance
- Note the male-to-male transmission, so it cannot be X-linked anything.
- So, it is likely autosomal dominant
- Also, an unusual situation where all children in generation-III are affected
- The risk of affected children for III-4 is 50%



Decision tree for determining mode of inheritance

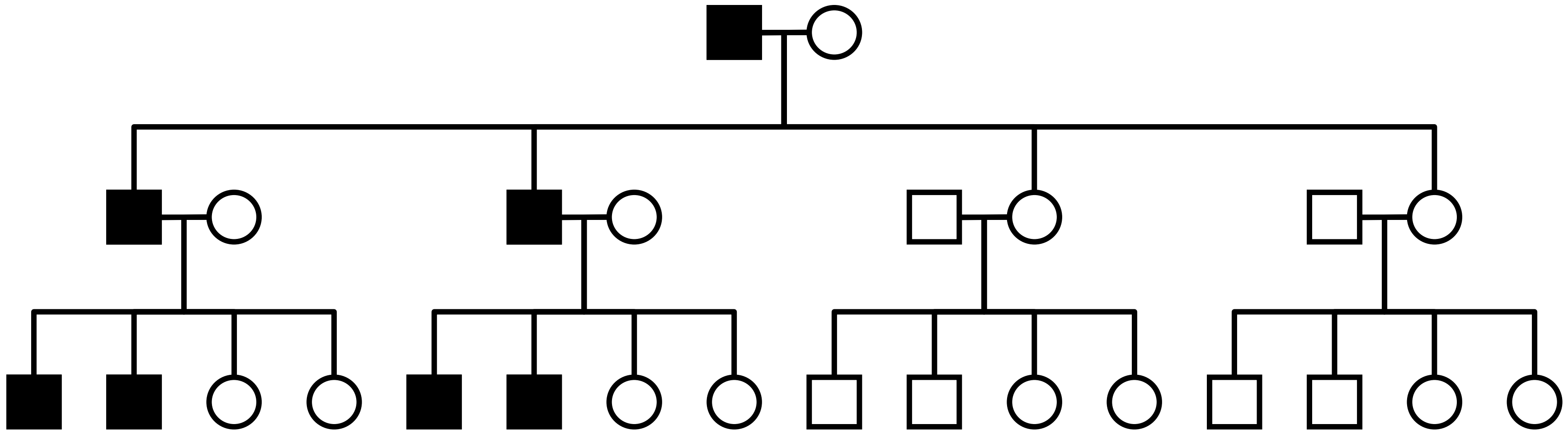
Use it every time for pedigree identification



Y-linked Inheritance

- Only males will show the trait.
- Genes on Y are primarily involved in spermatogenesis
- Therefore, mutations in these usually cause sterility and are not passed on.
  - Genetic lethality (the person may be relatively unaffected but unable to procreate)
- **Examples of Y-linked traits**
  - Various mutations in the SRY genes
  - H-Y histocompatibility antigen
  - Hairy ears

Y-Linked Inheritance: Male to Male Transmission



- *Y Linked Inheritance*
  - **Only Males affected**
  - Males transmit the trait to **all their sons** but none of their daughters
  - Main Point: If a trait is controlled by Y chromosome, it must have come from the father, as the mother is XX.

Unexpected phenotypes in Mendelian disorders as discussed in the pre-videos for this session

Students should be able to differentiate between the following “high-yield” terms:

- Reduced penetrance Vs variable expressivity
- Allelic heterogeneity Vs locus heterogeneity
- New mutation Vs germline mosaicism
- Compound heterozygote (affected) Vs homozygous affected

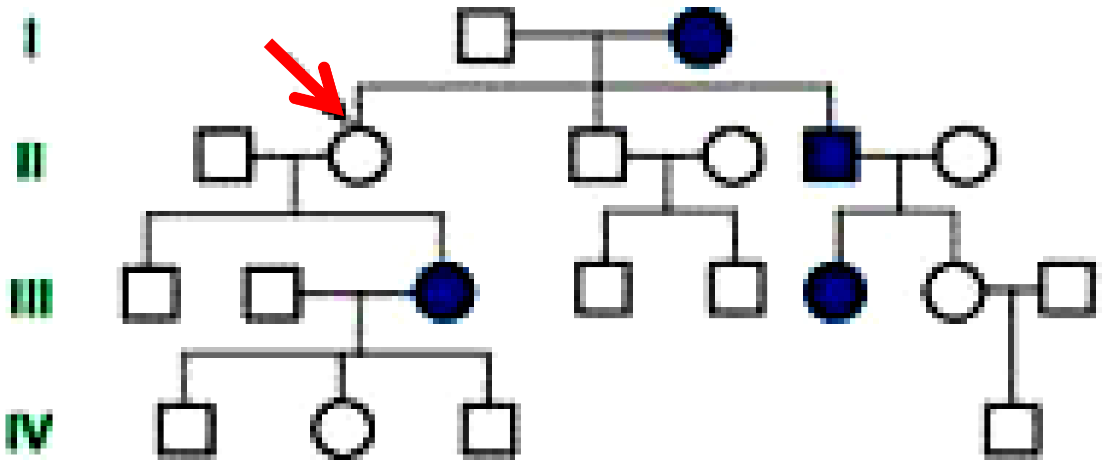
Important : From GNET DLA

Clicker Question Next!

Please prepare to answer
the clicker question
on the next slide.

A scientist studying a family segregating a known autosomal dominant genetic disorder constructs the pedigree shown. Which of the following best explains the phenotype in II-2 (red arrow)?

A.	Incomplete penetrance	20%
B.	Variable expression	20%
C.	Pleiotropy	20%
D.	Anticipation	20%
E.	Delayed age of onset	20%



72

From GNET DLA SOM.MKIII.BPM1.1.FTM.3.GNET.0052

A family with many members affected with an autosomal dominant disorder is given. The penetrance of the disorder is 75%. What is the risk of recurrence in the offspring of II-3 and II-4?

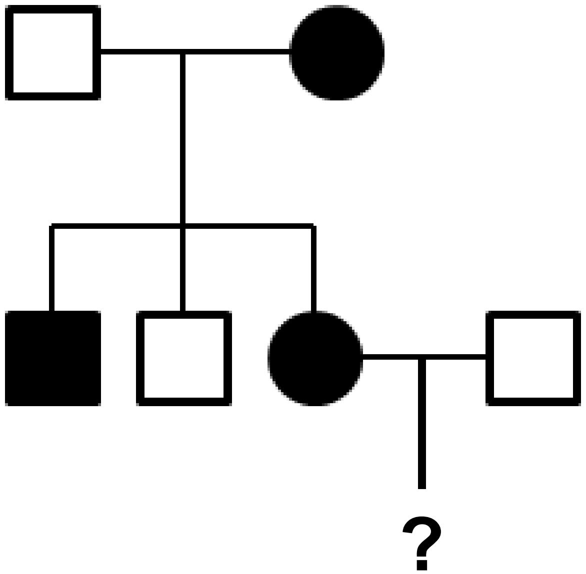
A. 50% 20%

B. 25% 20%

C. 37.5% 20%

D. 100% 20%

E. 0% 20%

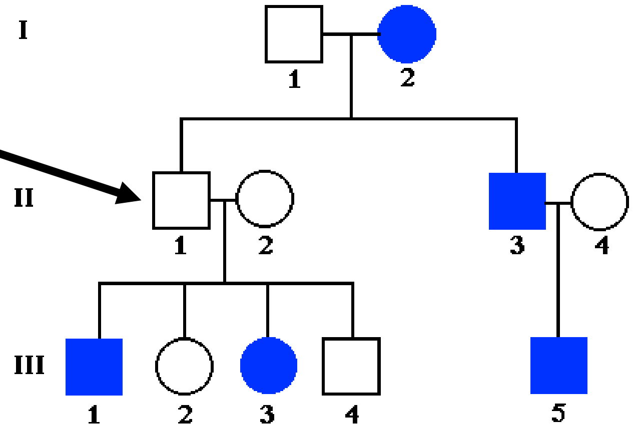


(50 x 75) / 100 = 37.5%

Reduced penetrance

From GNET DLA

Incomplete Penetrance



Incomplete, or reduced penetrance may complicate an autosomal dominant pedigree.

- Male to male transmission (II-3 to III-5)
- But III-1 and III-3 do not have an affected parent
- II-1 **must have inherited the mutant allele** from I-2 and has passed on the mutant allele to III-1 and III-3
- II-1 most probably has the disease genotype but does not display the disease phenotype: Example of **Non-penetrance**
- A disorder is said to be **fully penetrant**, if all the people carrying the mutation, express the phenotypic manifestations of the disorder

Clicker Question Next!

Please prepare to answer
the clicker question
on the next slide.

Genetic analysis of two individuals with Neurofibromatosis reveals the following results: In one patient, a point mutation (missense) in the *NF1* gene is identified and in the second patient a nonsense mutation in the *NF1* gene has been identified. What genetic principle is highlighted in this scenario?

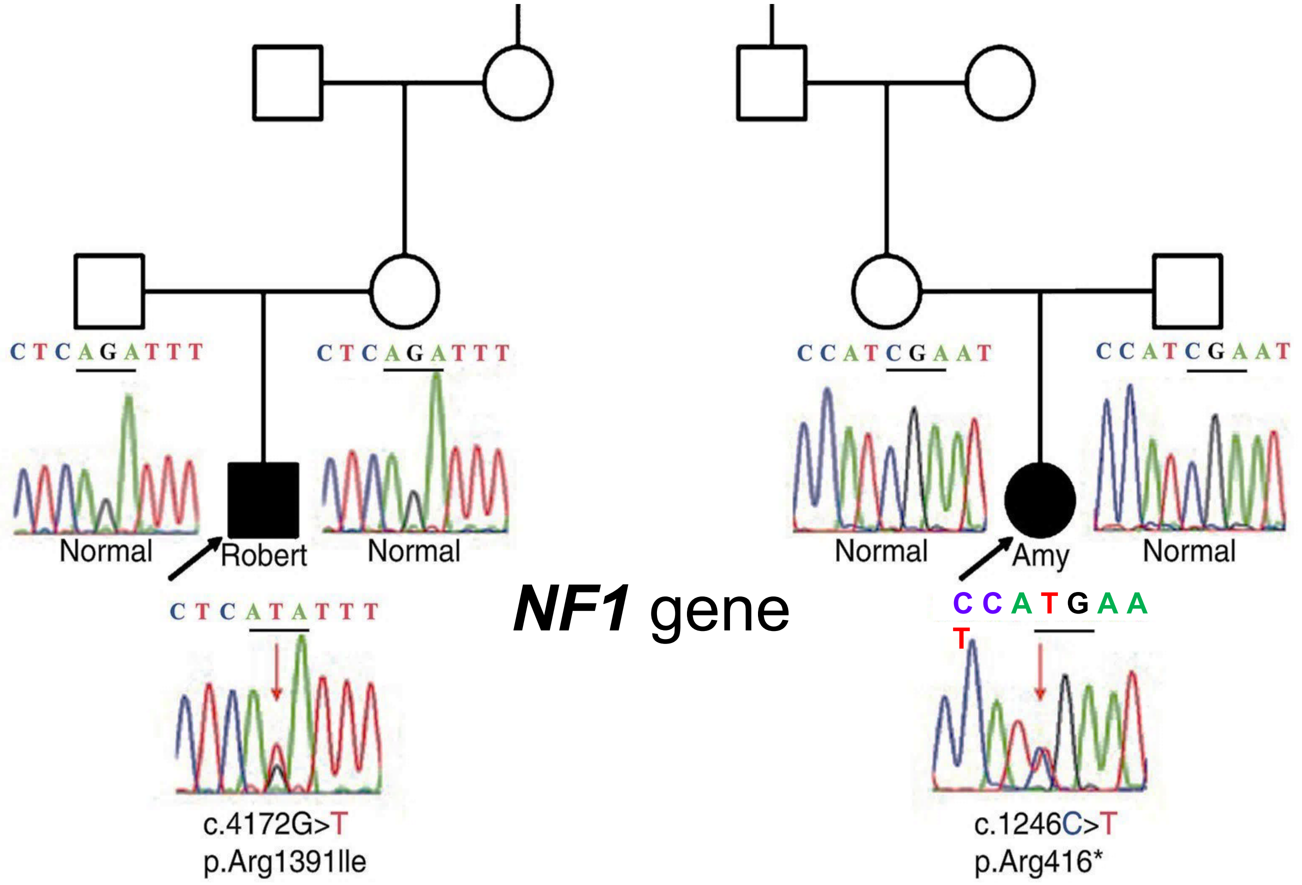
A.	Pleiotropy	20%
B.	Variable expression	20%
C.	Reduced penetrance	20%
D.	Locus heterogeneity	20%
E.	Allelic heterogeneity	20%

From GNET DLA

72

Neurofibromatosis

- Different types of mutations of the same gene (NF-1 gene)
  - Substitution mutation (Robert) in **NF1**
  - Nonsense mutation (Amy) in **NF1**



Human Genetics and Genomics, Fourth Edition. Bruce R. Korf and Mira B. Irons.

Figure 10.5 Sequencing results for Robert (left) and Amy (right), showing two *de novo* and distinct *NF1* mutations.

Clicker Question Next!

Please prepare to answer
the clicker question
on the next slide.

Charcot-Marie-Tooth disease has been identified in two unrelated patients. In one patient, a duplication of the *PMP22* gene is identified and in the second patient a mutation in the *MPZ* gene has been identified. What is the genetic principle highlighted in this scenario?

A.	Pleiotropy	20%
B.	Variable expression	20%
C.	Reduced penetrance	20%
D.	Locus heterogeneity	20%
E.	Allelic heterogeneity	20%

From GNET DLA

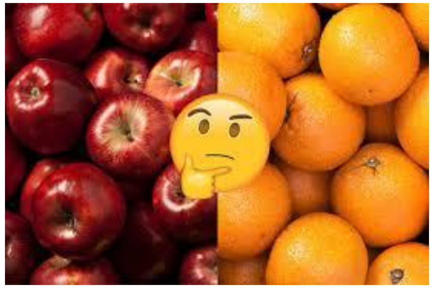
72

Locus heterogeneity

- Mutations at **different loci** that cause the same disease phenotype. Mutations of **different genes** cause the **same disease phenotype**.

- Osteogenesis imperfecta: Defect in collagen
  - Mutations of chromosome 17 (***COL1A1* gene**) or chromosome 7 (***COL1A2* gene**) lead to osteogenesis imperfecta

- Many disorders demonstrate locus heterogeneity (some examples):
  - SCID (AR and X-linked)
  - Familial hypercholesterolemia (LDLR or APOB)
  - Sensorineural hearing impairment
  - Charcot Marie Tooth disease (AD, AR, or X-linked)



Clicker Question Next!

Please prepare to answer
the clicker question
on the next slide.

Two siblings both have Neurofibromatosis due to the same inherited mutation. One sibling has several café au lait spots only. The other sibling has hundreds of neurofibromas, Lisch nodules of the eye, and has had surgery to remove tumors that were overly uncomfortable. Which genetic principle is best illustrated in this family?

A. New mutation 20%
B. Variable expression 20%
C. Reduced penetrance 20%
D. Locus heterogeneity 20%
E. Allelic heterogeneity 20%

From GNET DLA

72

Variable Expression

- In individuals who have inherited the same mutant allele, some individuals are severely affected, and others are mildly affected

- Various reasons:
  1. Random chance
  2. Other genetic factors (modifier loci) or sex influence
  3. Type of mutation: Marfan syndrome
  4. Environmental exposure ....

- **Some other examples: (students please → *read below on your own*)**

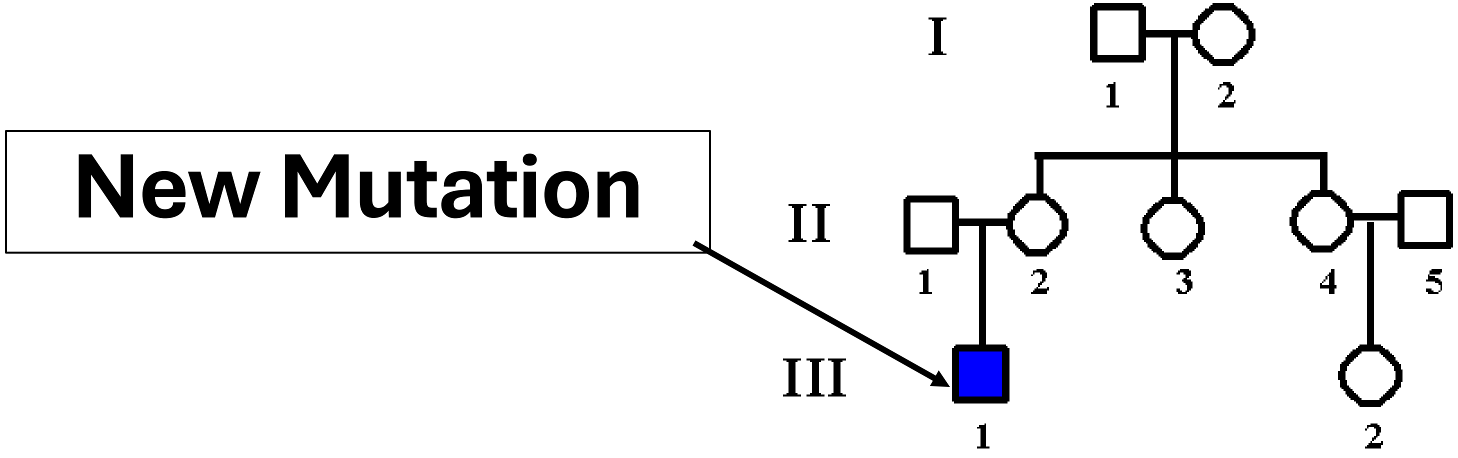
- Hemochromatosis – is an iron overload disorder (autosomal recessive disorder)
  - Hemochromatosis is more severe in males, less severe in females. Premenopausal females will menstruate and lose iron
  - Variable expression due to sex differences

- Xeroderma pigmentosum (autosomal recessive) – more severe in individuals exposed more frequently to environmental UV radiation
  - First child would be expected to be more severely affected than the second child
  - Parents might know to keep the second child out of the sun; variable expression due to environment

- Cystic fibrosis: two patients with the same exact mutation profile might present with very different clinical pictures. Could be different genetic modifiers/background, or different environment exposure.

- Some embryonic devolvement disorders (syndromic cleft lip), might be more severe in one child than in another, but caused by the same mutation – some of this might be due to random chance.

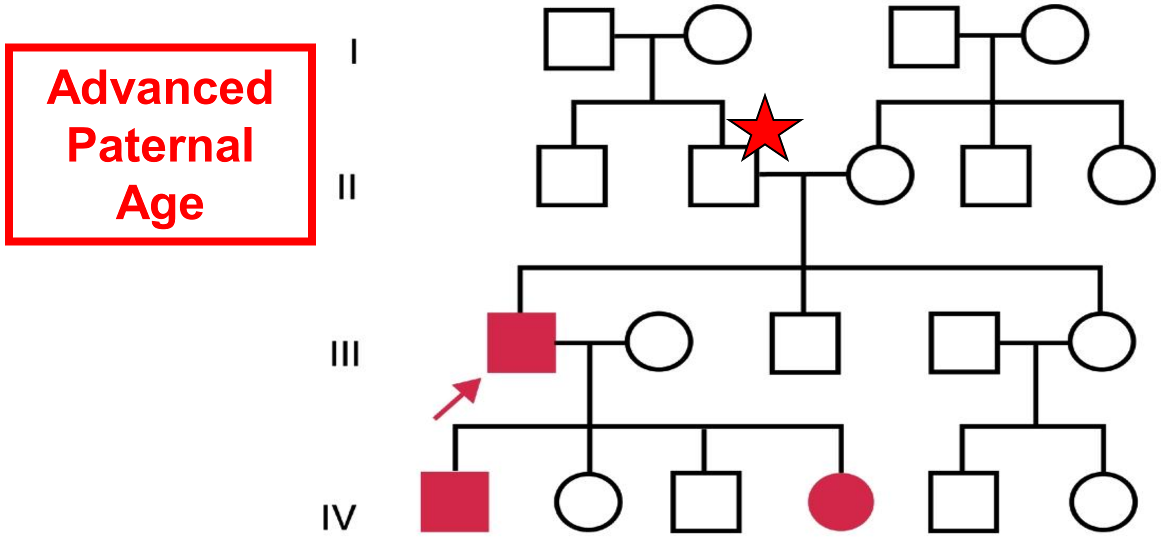
Many more examples will become apparent to you over the years



- The mutation is transmitted from an unaffected parent to an affected offspring
- In some cases, increased age of the father is observed
- There is NO family history of the disease
- Most frequent examples due to *hot spot* for mutation:
  1. Neurofibromatosis (*NF1*) – refer Chapter 10 (Korf)
  2. Achondroplasia (*FGFR3*)
  3. Duchenne muscular dystrophy (Dystrophin gene) – About 33% (1/3) of patients have a new mutation
- Other examples: Osteogenesis imperfecta, Marfan syndrome
- Typically, the risk of recurrence for the couple is low, but parents are never told their recurrence risk is same as population risk
  - Because there is chance that a parent is a germline mosaic.
- Counseling regarding recurrence is a challenge and is difficult to predict

From GNET DLA

Paternal Age & *de novo* Autosomal Dominant Mutations



<u>New</u> autosomal dominant mutations tend to occur more often in older fathers. Particularly men who can have a long reproductive life.

The mutation is not identified in somatic cells of II-2; Present only in the germline

Remember this mutation is occurring during gamete formation. It is spontaneous, (that is it is new), not in the parents original genome.

Effect of Paternal Age on the occurrence of new Autosomal dominant mutants

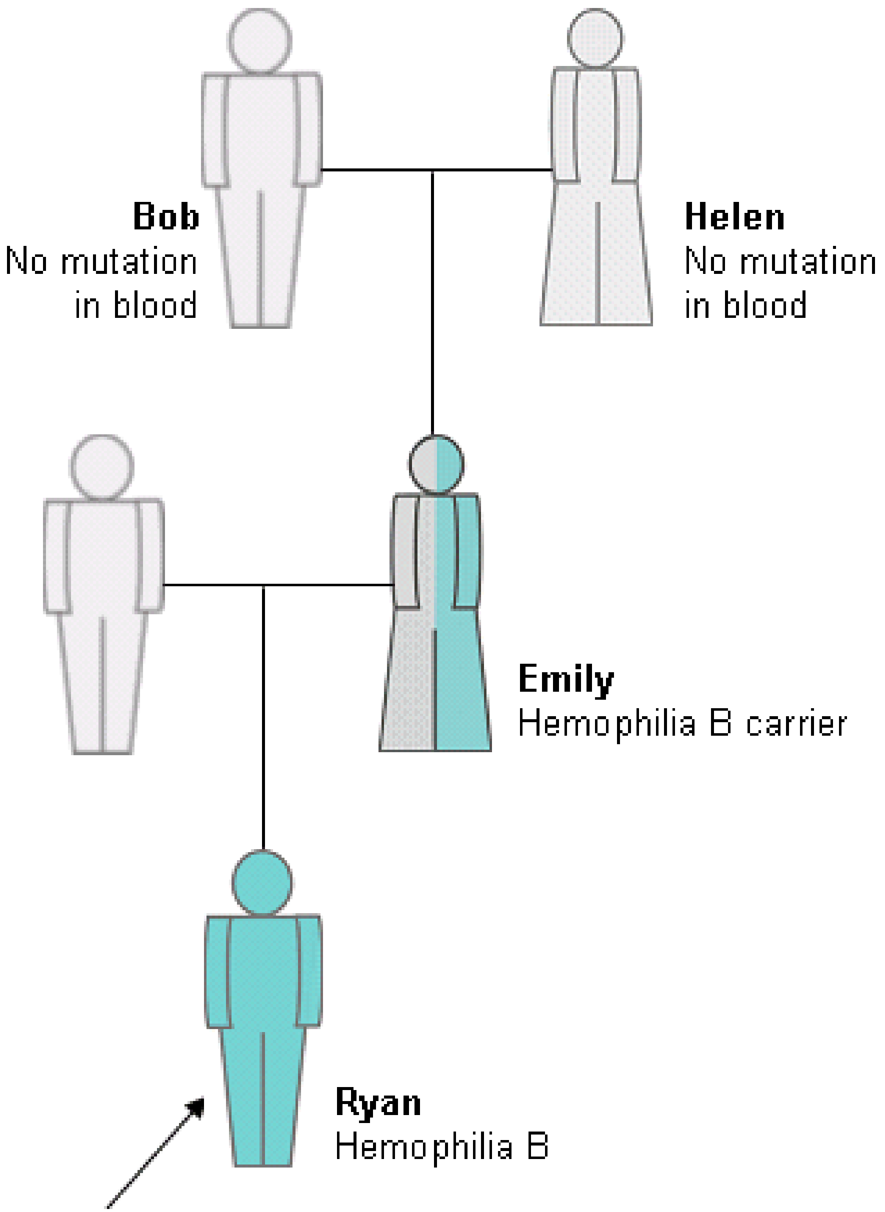
- ***The average age for fathers with*** children who have *new Autosomal Dominant mutations* is greater than the average age of the father in the general population
- Observed in several conditions
  - Average age of father in population **~30 - 31 yrs**
  - Average age of father of Achondroplasia **~36.1 yrs**
  - Average age of father of Marfan syndrome **~36.6 yrs**

  - About ½ of all *NF1* cases are *de-novo*, and risk increases with age of the father

- **Genetic advice is difficult; because there is nothing specific to advise on.**

- **Genetic Basis**: spermatogonia continually divide & sperm cells from older fathers may contain replication error mutations (point mutations)

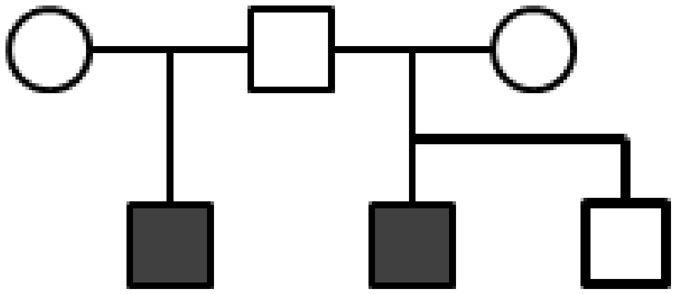
New mutation in X-linked recessive disorders



Possible explanations for Emily's hemophilia B carrier status:

- Spontaneous *F9* mutation
  - In single sperm from Bob
  - In single ovum from Helen
  - In fertilized egg, occurring at or soon after Emily's conception
- Germline mosaicism (with or without somatic mosaicism) for *F9* mutation in either Bob or Helen

Germline mosaicism



O.I.

O.I.

Same pathogenic variant as the half-brother

- A healthy man has two children affected with osteogenesis imperfecta with different partners. He also has another child that is normal.
- Genetic testing shows the same mutation in both affected children, but all parents test normal.
  - No parent has the mutation in their peripheral cells (somatic/ blood cells).
- May be due to **mosaicism** in the **germline (gonadal)** in the father
- The mutation is present in a proportion of the germline cells
- **Presence of two affected children suggests germline mosaicism (compare to new mutation, where there is a single child with a disorder and no family history of the disorder**)

Clicker Question Next!

Please prepare to answer
the clicker question
on the next slide.

From GNET DLA

A 23-year-old pregnant woman comes to the physician for counseling because her 60-year-old father was just diagnosed with Huntington disease. What is the risk that that she might transmit the pathogenic triplet repeat expansion to her child?

Her probability of having the pathogenic variant is ½ (autosomal dominant - 50% risk of transmission to offspring)

The probability of her passing the pathogenic variant to her child is ½

A.	2/3	20%
B.	½	20%
C.	¼	20%
D.	1/8	20%
E.	1	20%

71

SOM.MKIII.BPM1.1.FTM.3.GNET.0053

Delayed age of onset

- Individuals inherit the disease-causing variant at birth, but do not manifest the phenotype until later in life.
- Many examples illustrate this principle, below are a few:
- **Huntington disease** (autosomal dominant; trinucleotide repeat expansion)
  - Often manifests after a person has children (say in a person's 40s)
  - Progressive dementia, loss of motor control (chorea)
- **Hemochromatosis (iron overload disorder)** – typically presents by 30-40 years of age in males; usually later onset in females (because of menstruation)
- **Familial breast cancer (*BRCA-1* or *BRCA-2* mutation)**
  - **Increased BrCa likelihood as person ages**

- Delayed age of onset has to be kept in mind when writing pedigrees for such disorders & also when predicting the risk to the offspring

A study of the *CFTR* mutation in two separate families, revealed a codon deletion in one family and a missense mutation in the second family. A marriage occurred between members of these two families, and a child was born with CF. Which term describes this patient?

A. Incomplete penetrance 14%
B. Locus heterogeneity 14%
C. Anticipation 14%
D. Digenic inheritance 14%
E. Compound heterozygote 14%
F. Pleiotropic effect 14%
G. Heteroplasmy 14%

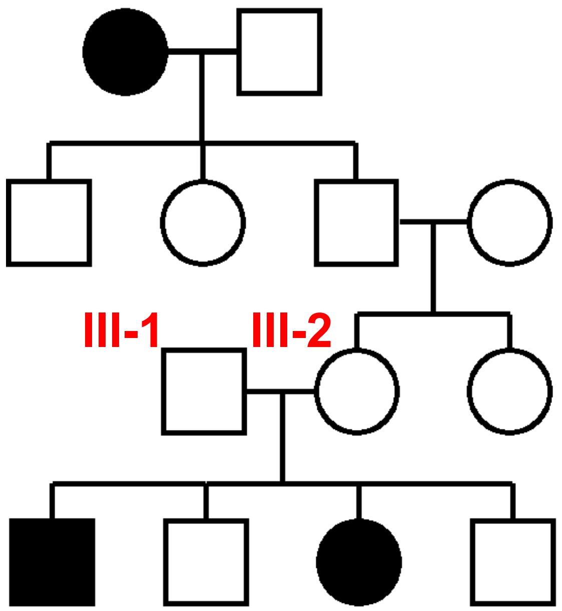
Compound heterozygote is due to allelic heterogeneity of the CFTR gene

72

Pleiotropy

- A disease-causing mutation (single gene) affects more than one organ system
- **Marfan syndrome (autosomal dominant)**
  - Mutation in the **fibrillin-1 gene**
  - Skeletal abnormalities (arachnodactyly, long limbs, pectus excavatum)
  - Hypermobile joints
  - Ocular abnormalities (myopia, lens dislocation)
  - Cardiovascular disease (mitral valve prolapse, aortic aneurysm)
- **Osteogenesis imperfecta** (brittle bones, blue sclera) is due to a mutation in a gene encoding collagen.
- **Cystic fibrosis**: Respiratory infections and malabsorption

A scientist is studying a family from a region where PKU is found at relatively high prevalence. The pedigree shows a family with several members affected with phenylketonuria. What would be genotypes for III-1 and III-2?



	III-1	III-2	
A.	AA,	AA	13%
B.	Aa,	aa	13%
C.	aA,	aA	13%
D.	a,	AA	13%
E.	A,	Aa	13%
F.	aY,	Aa	13%
G.	aa,	aa	13%
H.	AA,	aa	13%

72

PKU is autosomal recessive; Risk to their next child is 25%